Shipping BIBFRAME

Using Bounded Descriptions

Kevin Ford (Library of Congress)
Ed Summers (Stanford University)

Overview

1. The Problem
2. Concise Bounded Descriptions (CBD) to the rescue?
3. How are Bounded Descriptions used at the Library of Congress?
4. How are Bounded Descriptions used in the Blue Core Project?
5. Existing work on validating BIBFRAME.
6. BIBFRAME JSON validation.

The Problem

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<bf:Work rdf:about="http://id.loc.gov/resources/works/12376507">
  <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/Text"/>
  <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/Monograph"/>
  <bf:language rdf:resource="http://id.loc.gov/vocabulary/languages/eng"/>
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  <bf:subject rdf:resource="http://id.loc.gov/authorities/subjects/sh85073009"/>
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```

The Problem

```
<bf:Work rdf:about="http://id.loc.gov/resources/works/12376507">
  <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/Text"/>
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  <bf:subject rdf:resource="http://id.loc.gov/authorities/subjects/sh85073009"/>
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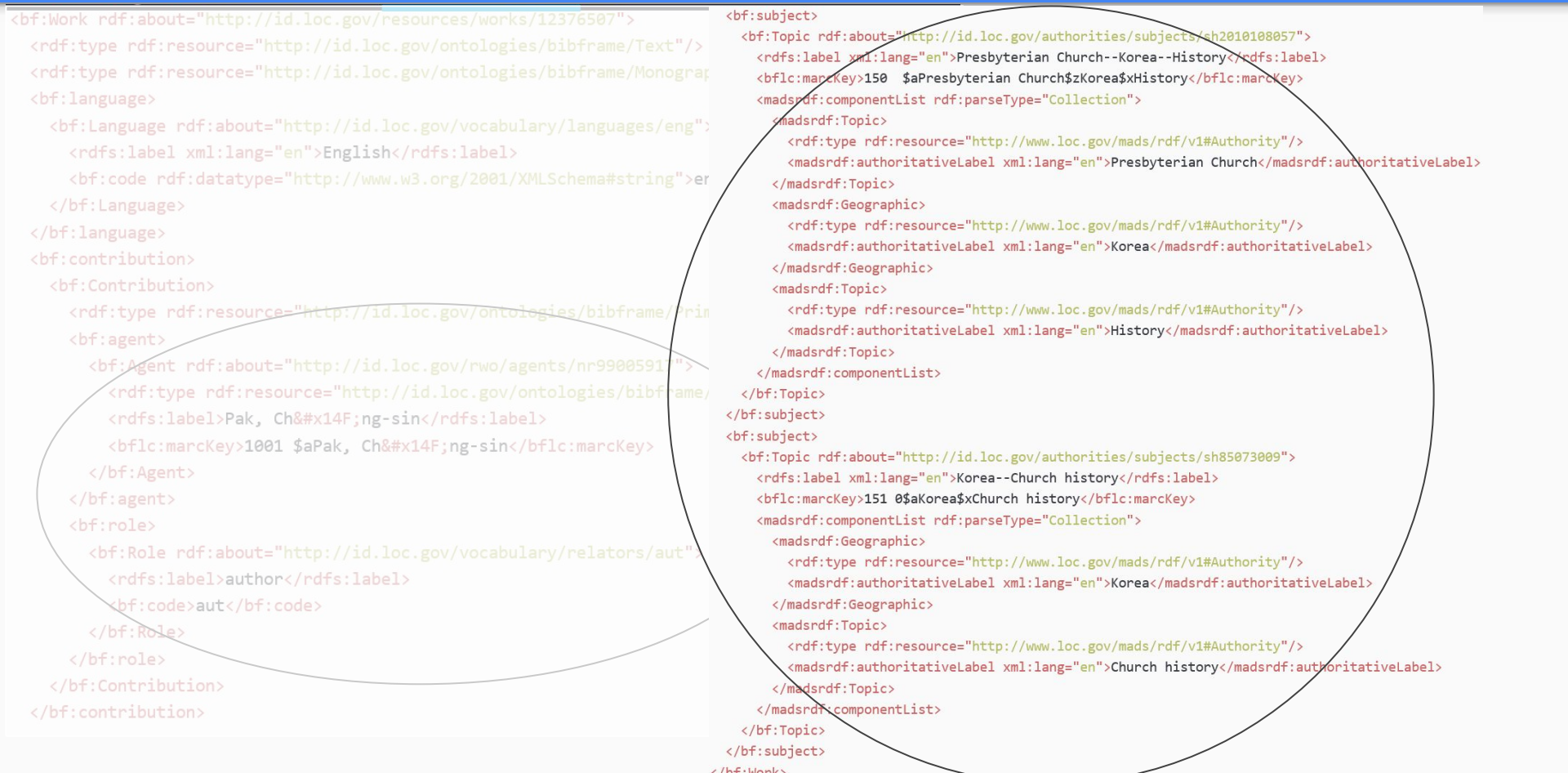
The Problem

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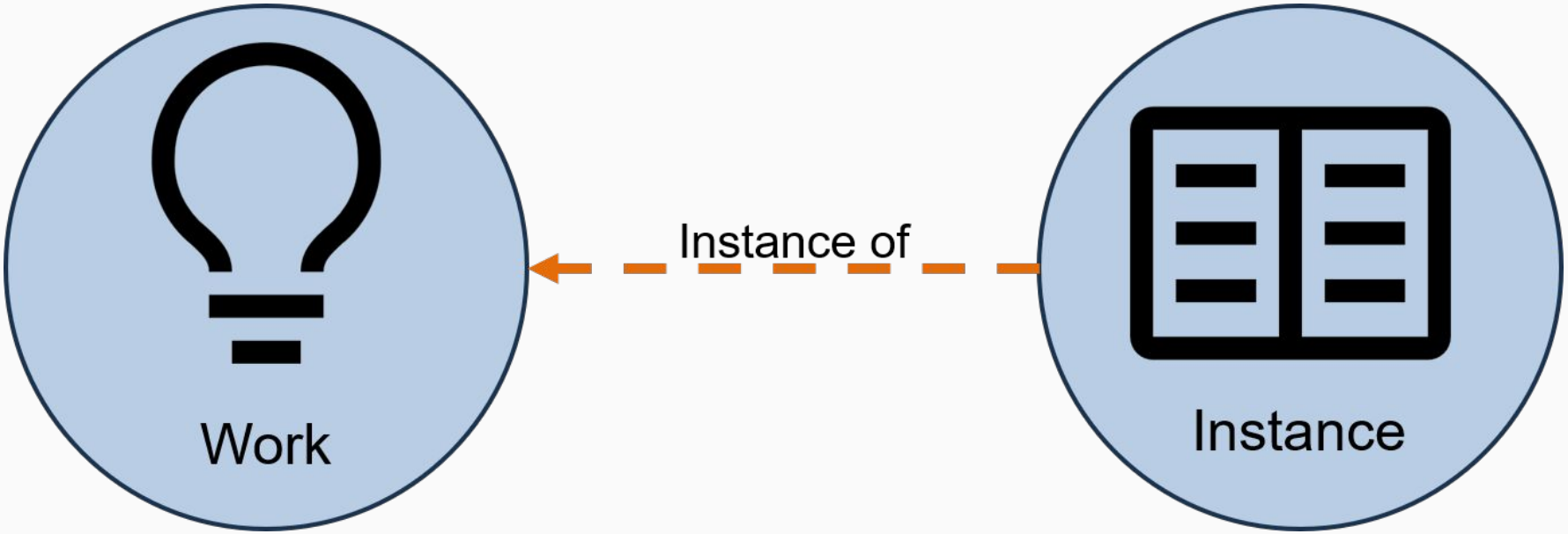
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  <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/Monograph"/>
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      <rdfs:label xml:lang="en">English</rdfs:label>
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The Problem


<bf:Work rdf:about="http://id.loc.gov/resources/works/12376507">
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The Problem



The Problem

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Dewey: 280/.4/09519 221
Author: Pak, Chŏng-sin.
Title: Protestantism and politics in Korea
Subject: Christianity and politics--Korea--History
Subject: Presbyterian Church--Korea--History
Subject: Korea--Church history

Instance of

LCCN: 2001027688
ISBN: 0295981490 (alk. paper)
Publication: Seattle, WA : University of Washington Press, c2003.
Description: axi, 316 p. ; 24 cm.
Note: Includes bibliographical references (p. 273-301) and index.
Medium: unmediated
Carrier: volume

The Problem

Instance of

```
<bf:Work rdf:about="http://id.loc.gov/resources/works/12376507">
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  <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/Monograph"/>
  <bf:language rdf:resource="http://id.loc.gov/vocabulary/languages/eng"/>
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      <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/PrimaryContribution"/>
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      <bf:role rdf:resource="http://id.loc.gov/vocabulary/relators/aut"/>
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The Problem

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</bf:Work>
```

What is a CBD?



Member Submission

CBD - Concise Bounded Description

W3C Member Submission 3 June 2005

This version:
http://www.w3.org/submissions/2005/SUBM-CBD-20050603/
Latest version:
http://www.w3.org/submissions/CBD/
Previous version:
http://www.w3.org/submissions/2004/SUBM-CBD-20040930/
Author:
Patrick Stickler, Nokia, patrick.stickler@nokia.com

Copyright Nokia 2003-2004.

What is a CBD?

- A *concise bounded description* of a resource [is represented] in terms of **an RDF graph,** as a general and broadly optimal unit of specific knowledge about that resource to be utilized by, and/or interchanged between, semantic web agents.

- Given a particular node in a particular RDF graph, a *concise bounded description* is a subgraph consisting of those statements which together constitute a focused body of knowledge about the resource denoted by that particular node.

What is a CBD?

Definition

Given a particular node (the starting node) in a particular RDF graph (the source graph), a subgraph of that particular graph, taken to comprise a *concise bounded description* of the resource denoted by the starting node, can be identified as follows:

1. Include in the subgraph all statements in the source graph where the subject of the statement is the starting node;
2. Recursively, for all statements identified in the subgraph thus far having a blank node object, include in the subgraph all statements in the source graph where the subject of the statement is the blank node in question and which are not already included in the subgraph.
3. Recursively, for all statements included in the subgraph thus far, for all reifications of each statement in the source graph, include the *concise bounded description* beginning from the rdf:Statement node of each reification.

This results in a subgraph where the object nodes are either URI references, literals, or blank nodes not serving as the subject of any statement in the graph.

What is a CBD?

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3. Recursively, for all statements included in the subgraph thus far, for all reifications of each statement in the source graph, include the *concise bounded description* beginning from the rdf:Statement node of each reification.

This results in a subgraph where the object nodes are either URI references, literals, or blank nodes not serving as the subject of any statement in the graph.

BIBFRAME Bounded Descriptions at LC

BF Bounded Description includes:

- At least one Instance and one Work
    - All triples
- Any referenced Items
    - All triples
- For Hubs, Agents, Subjects, Carrier, Media, etc.
    - Not all triples
    - Minimally labels (non-latin also) and often codes
    - MARC Keys where available
- Depth of introspection may vary depending on property/class.

BIBFRAME Bounded Descriptions at LC

```
<bf:contribution>
 −<bf:Contribution>
    <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/PrimaryContribution"/>
  −<bf:agent>
    −<bf:Agent rdf:about="http://id.loc.gov/rwo/agents/nr99005917">
       <rdf:type rdf:resource="http://id.loc.gov/ontologies/bibframe/Person"/>
       <rdfs:label>Pak, Chŏng-sin</rdfs:label>
       <bflc:marcKey>1001 $aPak, Chŏng-sin</bflc:marcKey>
       <rdfs:label xml:lang="ko">박 정신</rdfs:label>
       <bflc:marcKey xml:lang="ko">4001 $a박 정신$7(bcp47)ko-hang</bflc:marcKey>
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   </bf:role>
 </bf:Contribution>
</bf:contribution>
```

BIBFRAME Bounded Descriptions at LC

Uses:

- Data package loaded to Marva Editor
- Data package passed through BIBFRAME-to-MARC Conversion

Example URL: https://id.loc.gov/resources/instances/12376507.bd.rdf

BIBFRAME Bounded Descriptions at LC

- BIBFRAME Primer: https://bibframe.org/primer
- Specifically the “RDF in BIBFRAME” section: https://bibframe.org/docs/view/documentation-bf-primer/rdf-in-bibframe/index.md#rdf-in-bibframe

Pattern: https://id.loc.gov/resources/instances/{identifier}.bd.[rdf|jsonld|nt]

Example URL: https://id.loc.gov/resources/instances/12376507.bd.rdf

BIBFRAME Bounded Descriptions at LC

- BIBFRAME Primer: https://bibframe.org/primer
- Specifically the “RDF in BIBFRAME” section: https://bibframe.org/docs/view/documentation-bf-primer/rdf-in-bibframe/index.md#rdf-in-bibframe

Pattern: https://id.loc.gov/resources/instances/{identifier}.bd.[rdf|jsonld|nt]

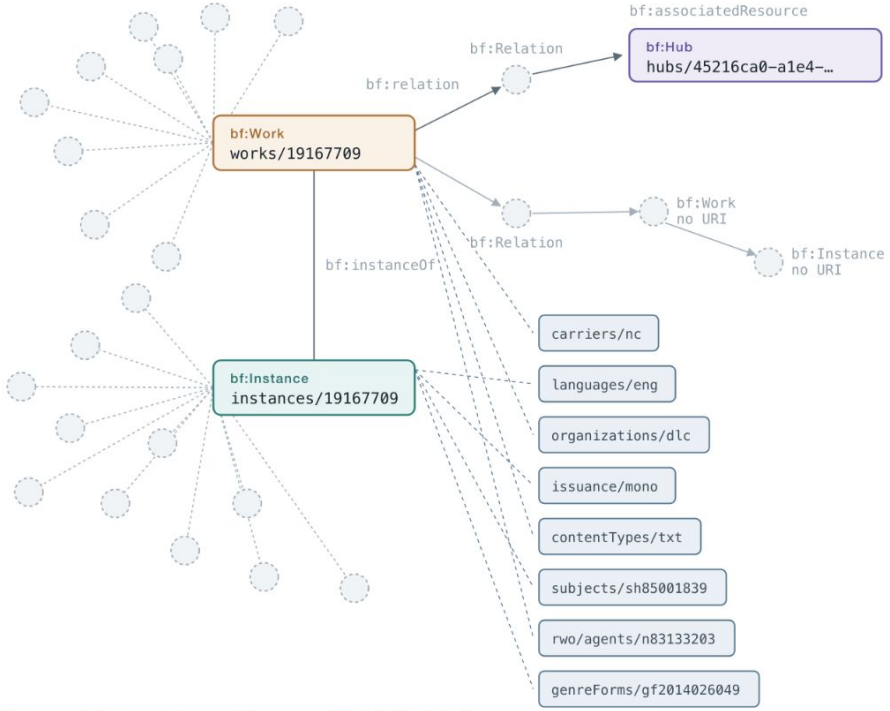
Example URL: https://id.loc.gov/resources/instances/12376507.bd.rdf

Use of CBD in Blue Core

https://bluecore.info

A CBD arrives as one undivided graph

Nothing in the triples says where one resource ends and the next begins.



We parse id.loc.gov/resources/instances/19167709.cbd.rdf
into 335 triples, 86 URIs and 48 blank nodes.

+ 21 more

BLUE CORE DATABASE

WORKS
no rows yet

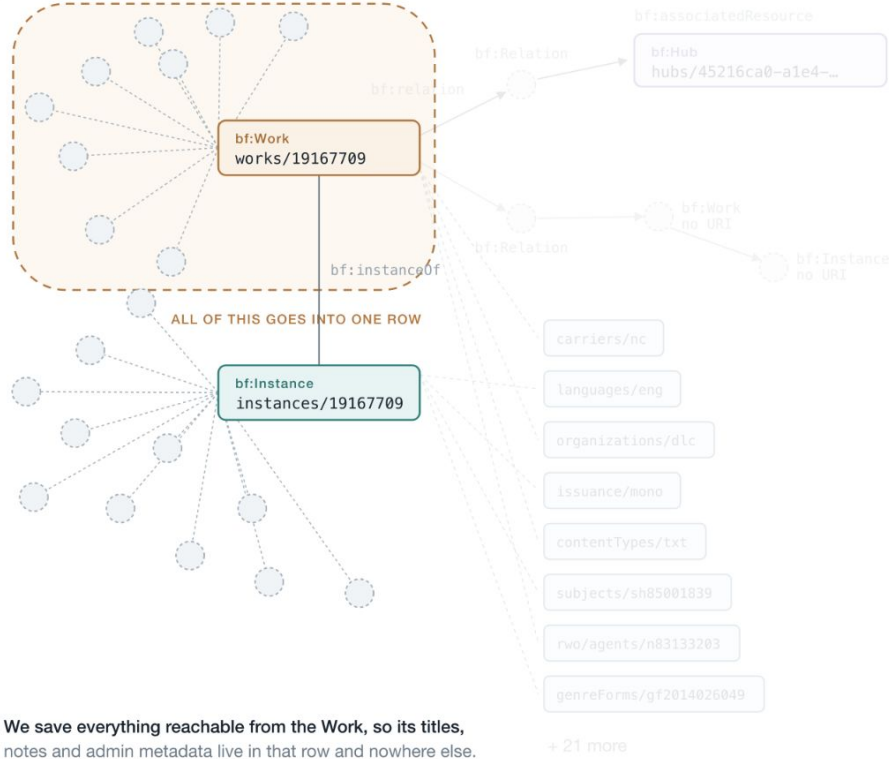
INSTANCES
no rows yet

HUBS
no rows yet

OTHER RESOURCES
no rows yet

A resource is a subgraph, not a node

Nine blank nodes hang off this Work, and more hang off those. We keep the whole cluster together.



BLUE CORE DATABASE

WORKS
bcld.info/works/{uuid}
1

INSTANCES
no rows yet

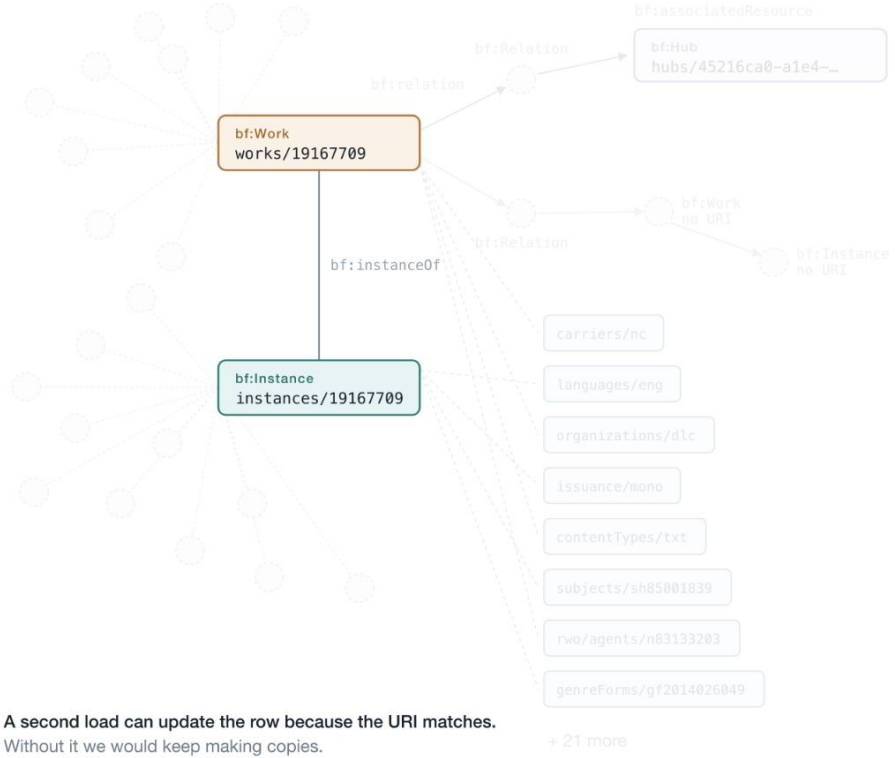
HUBS
no rows yet

OTHER RESOURCES
no rows yet

We save everything reachable from the Work, so its titles,
notes and admin metadata live in that row and nowhere else.

The record is about a Work and an Instance

Both carry a URI, so we recognise them the next time this record loads and update the rows we have.



BLUE CORE DATABASE

WORKS
bcld.info/works/{uuid}
1

INSTANCES
bcld.info/instances/{uuid}
1

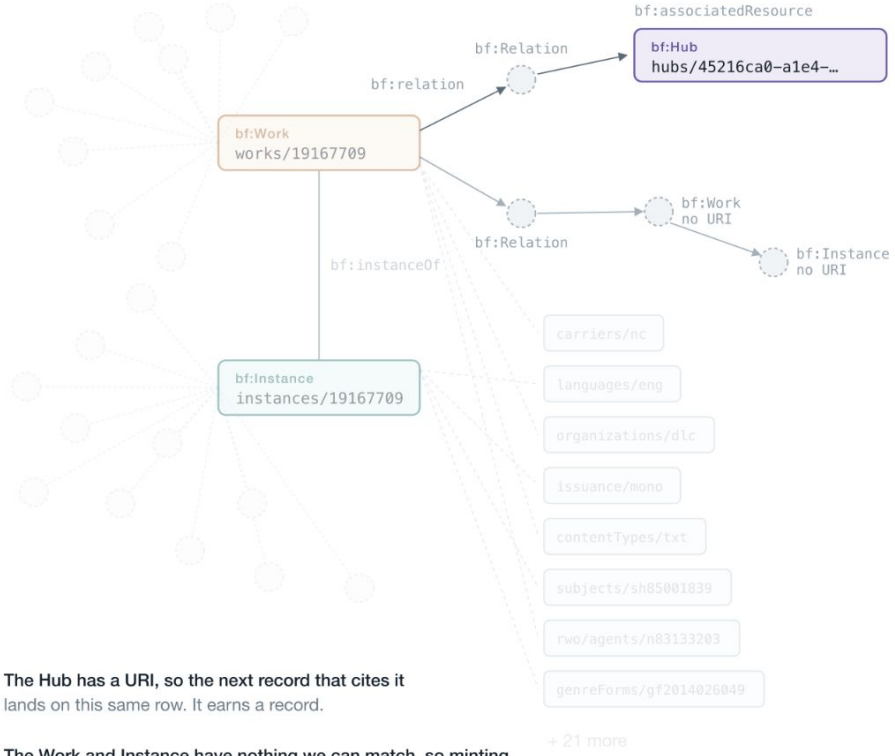
HUBS
no rows yet

OTHER RESOURCES
no rows yet

A second load can update the row because the URI matches.
Without it we would keep making copies.

Is the far end of a relation a resource, or just a mention?

Both branches look identical in the data, and answering wrong grows the database on every load.



The Hub has a URI, so the next record that cites it lands on this same row. It earns a record.

The Work and Instance have nothing we can match, so minting a URI invents a new pair on every load. They stay mentions.

BLUE CORE DATABASE

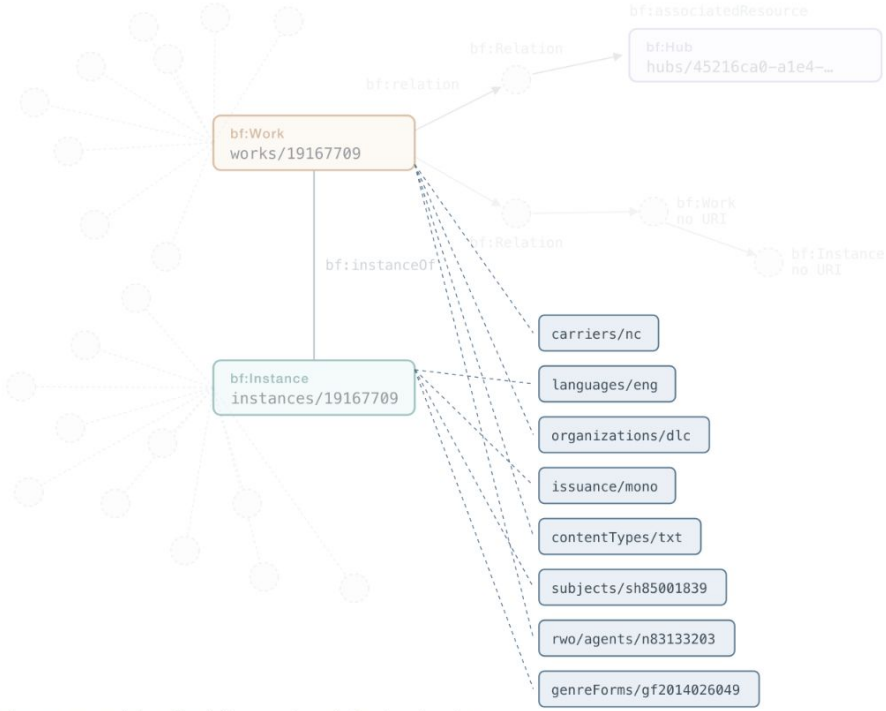
WORKS bcld.info/works/{uuid}	1
INSTANCES bcld.info/instances/{uuid}	1
HUBS bcld.info/hubs/{uuid}	1

OTHER RESOURCES

no rows yet

The record points at 29 URIs it does not own

Carriers, languages, agents and subject headings. We store each one once and link it many-to-many.



+ 21 more

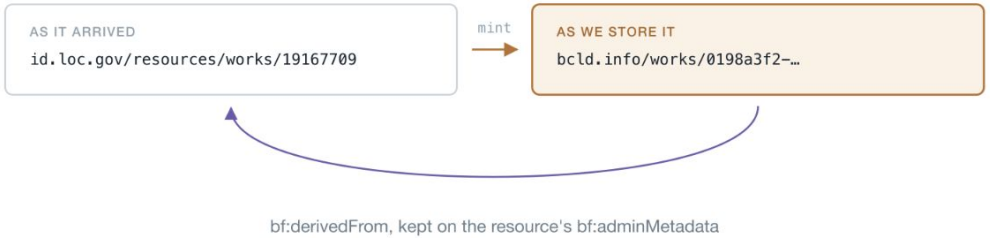
The next record that cites id.loc.gov/vocabulary/carriers/nc points at this same row.

BLUE CORE DATABASE

WORKS	bcld.info/works/{uuid}	1
INSTANCES	bcld.info/instances/{uuid}	1
HUBS	bcld.info/hubs/{uuid}	1
OTHER RESOURCES	id.loc.gov/… unchanged	29

Whose identifier is it?

We curate Works, Instances and Hubs. We do not curate the nodes we share with everyone else.



A re-harvest finds this record by where it came from
instead of creating a second one beside it.

NOT MINTED

`id.loc.gov/vocabulary/carriers/nc`
`id.loc.gov/authorities/subjects/sh85001839`
`id.loc.gov/rwo/agents/n83133203`

Other Resources keep LC's identifier.
Minting our own would fork the shared vocabulary into a private copy and cut the one link that lets records from anywhere meet on the same node.

MINTED

`bf:Work` `bf:Instance` `bf:Hub`

BIBFRAME Validation

BIBFRAME Validation at LC:

- Requires, minimally, the Work and Instance
  - Validating only the Work *or* the Instance results in incomplete view
- Validation performed during cataloging
- Validation is a custom, system-based routine

BIBFRAME Validation

BIBFRAME Validation and BIBFRAME Interoperability Group:

- Allows only Work or only Instance or both Work and Instance
- Still determining place in workflow
- Validation rules managed in standard way -> DCTAP
- Validation rules converted to standards-based RDF vocabulary -> SHACL
- Validation performed using readily available tooling -> pySHAC or Jena or …

BIG DCTAP Rules: https://github.com/bf-interop/DCTap

BIBFRAME JSON

Since JSON is such a popular format for Web APIs the Blue Core database stores BIBFRAME JSON-LD.

A given RDF graph can be represented in *many* different ways as JSON-LD, which can make JSON interoperability difficult.

The bibframe-json project provides a **JSON Schema** for CBD and resource centric BIBFRAME JSON, to help ensure that there is a consistent shape to build applications around.

INSTANCE

```
{
  "@context": "…/context/bibframe.jsonld",
  "@id": "https://bibframe.example/instances/0b46c016-…",
  "@type": ["Instance"],
  "title": [
    { "@type": ["Title"], "mainTitle": ["The Conceptual Structure of Reality"] }
  ],
  "identifiedBy": [{ "@type": ["Isbn"], "rdf:value": ["9783319071862"] }],
  "instanceOf": [
    {
      "@id": "https://bibframe.example/works/4be57b41-…",
      "@type": ["Work"],
      "title": [
        { "@type": ["Title"], "mainTitle": ["The Conceptual Structure of Reality"] }
      ],
      "hasInstance": ["https://bibframe.example/instances/0b46c016-…"]
    }
  ]
}
```

WORK

https://blue-core-lod.github.io/bibframe-json/

Want to learn more?

- BIBFRAME Primer: https://bibframe.org/docs/view/documentation-bf-primer/index.md
- BIBFRAME JSON Documentation: https://blue-core-lod.github.io/bibframe-json
- BIBFRAME Interoperability Group: https://www.bibframe.org/dctap/
- BIBFRAME discussion list: https://listserv.loc.gov/cgi-bin/wa?SUBED1=bibframe&A=1

Thank you!

Ed Summers (Stanford University)
Kevin Ford (Library of Congress)